Online Supplemental Materials

Supplemental 1. Interpreting the interaction between the average similarity effect and peer variance in the behavioral dynamics model

Change in discrete dependent behavior variables in a stochastic actor-oriented model is modeled by two stochastic subprocesses, modeling the rate of change and the direction of change, respectively (Snijders, Steglich, & Schweinberger, 2007). In particular, the second subprocess models the mechanisms that determine whether an actor, when the opportunity arises, increases or decreases their behavior level by one unit, or maintains their current level. This choice is modeled by a multinomial choice model (McFadden, 1974) that takes into account the behavior states s of all actors, the current network state x, and actor and dyadic covariates. This model depends on a so-called objective function, which for an actor i is given by

$$f_i(x, s) = \sum_k \beta_k e_{ik}(x, s).$$

The terms e_{ik} are effects representing the mechanisms for behavior change and the β_k are parameters indicating the strengths of these effects. The objective function can be interpreted as the actor's satisfaction with (or preference for) the result of the behavior change decision. Suppose for an actor i, the current behavior state is s_i^c. Then the probability that he or she selects s_i^n as the new behavior state is expressed in multinomial logit shape (McFadden, 1974),

$$p_i(s') = \exp\big(f_i(x, s')\big) \Big/ \sum_{s''} \exp\big(f_i(x, s'')\big)$$

where s' represents the behavioral dependent variable vector in which only the behavior state for actor i changes from s_i^c to s_i^n while the behavior states for all other actors do not change. The denominator sums over all possible next behavior states for actor i (current level, one unit

higher, one unit lower – except for actors with scores on the outer ends of the scale). If β_k equals 0, the corresponding effect plays no role in the behavior dynamics; if β_k is positive, then there will be a higher probability of moving into directions where the corresponding effect is higher, and the converse if β_k is negative.

Since our study seeks to explore the moderating role of peer variance on social influence, in this section, we zoom in on two effects: the average similarity effect, operationalizing social influence, and the peer variance effect. In particular, we explain how the interaction between these two effects can be visualized and interpreted. Since the current study focuses on stress as a discrete dependent behavior variable, we will set up the interpretation of the interaction between average similarity and peer variance in this context. However, the arguments generalize to any discrete dependent behavior variable in a stochastic actor-oriented model.

Average similarity and peer variance

The average similarity effect is defined by the average of the centered scores Sim_{ij} between an ego actor i and the other actors j who receive ties from the ego actor. Here, an ego indicates the actor who makes the decision to change or maintain his or her stress level, and an alter indicates a network member to whom this ego person sends an outgoing tie. The similarity score between two actors i and j is defined by

$$Sim_{ij} = 1 - \frac{|s_i - s_j|}{r_s}$$

where r_s represents the range of stress (the difference between the highest and lowest values of stress), which in our study is equal to 4 (stress ranges from 1 to 5). The lowest value of the stress similarity score is 0, for two individuals having two different extreme values of stress, while the highest value is 1, for two individuals having exactly the same stress values. The average similarity effect is defined as

$$avSim_i = \frac{1}{\sum_j x_{ij}} \sum_j x_{ij}(Sim_{ij} - \widehat{Sim})$$

where x_{ij} equals to 1 when the person i sends a tie to the person j, and 0 otherwise, where the sum is taken over all network members. The average similarity score is centered by subtracting the mean similarity score over all dyads at all time points $\widehat{Sim}$. When an actor i does not have any outgoing ties, their average similarity score is set to 0. The average similarity effect reflects a propensity of the actors to become similar or not in stress to their alters.

The peer variance effect (or alter variance) is defined as

$$altVar_i = \frac{1}{\sum_j x_{ij} - 1} \sum_j x_{ij} \, (s_j - \bar{s}_{alt(i)})^2 - Var^{obs}$$

where $\bar{s}_{alt(i)}$ indicates the average stress score of actor i's alters. Note here that the peer variance score for actors without outgoing ties or with only one outgoing tie is 0 by definition. This effect is centered around the variance of stress observed over all individuals at all time points Var^{obs}.

Interpreting the average similarity effect

The influence on stress an actor experiences from their social contacts, as operationalized by the average similarity effect, depends only on the proportion of alters whose stress levels are higher than their own stress level, not on how much larger or smaller the alters' values are. This can be explained as follows. For a specific actor i, let the numbers of their alters whose stress levels are lower than, equal to, or higher than their own stress level, be denoted by a, b, and c. Let the stress levels for low stress alters be denoted by s_1^l, s_2^l, …, s_a^l, for high stress alters s_1^h, s_2^h, …, s_c^h, and for equal stress alters s_1^e, s_2^e, …, s_b^e. Then the average similarity score between actor i and alters can be expressed as

$$avSim_{ij}(x,s) = \frac{1}{a+b+c}\left(\sum_{j=1}^{a}\left(1-\frac{|s_i - s_j^l|}{r_s} - \widehat{Sim}\right) + \sum_{j=1}^{b}\left(1-\frac{|s_i - s_j^e|}{r_s} - \widehat{Sim}\right) + \sum_{j=1}^{c}\left(1-\frac{|s_i - s_j^h|}{r_s} - \widehat{Sim}\right)\right)$$

$$= \frac{1}{a+b+c}\left(\sum_{j=1}^{a}\left(1-\frac{|s_i - s_j^l|}{r_s}\right) + \mathrm{b} + \sum_{j=1}^{c}\left(1-\frac{|s_i - s_j^h|}{r_s}\right)\right) - \widehat{Sim}$$

$$= \frac{1}{a+b+c}\left(a - \frac{as_i - \sum_{j=1}^{a} s_j^l}{r_s} + \mathrm{b} + c - \frac{\sum_{j=1}^{c} s_j^h - cs_i}{r_s}\right) - \widehat{Sim}$$

$$= \frac{1}{a+b+c}(a + \mathrm{b} + c - \frac{(a-c)s_i + \sum_{j=1}^{c} s_j^h - \sum_{j=1}^{a} s_j^l}{r_s}) - \widehat{Sim}$$

Since the behavior choices are modeled by multinomial logistic regression, the model parameters can be interpreted as contributions to log-probabilities. In particular, the log odds of actor i increasing their current stress state s_i^c to one level higher $(s_i^c + 1)$ instead of staying the same can be expressed as

$$log\left(\frac{p\left(s_i^c \; to \; (s_i^c + 1)\right)}{p(s_i^c \; to \; s_i^c)}\right) = f(x, s') - f(x, s)$$

Therefore, the value of the odds, i.e., the probability of increasing stress level by one unit over the probability of maintaining the same stress level, is $exp(f(x,s') - f(x,s))$.

In assessing the contribution of the average similarity effect to these log odds, we note that by increasing their stress level, an actor i gets more dissimilar to the $a + b$ friends now scoring lower, while getting more similar to the c friends who previously scored higher. The average similarity score after i increases stress by one unit would be

$$\frac{1}{a+b+c}\left(\sum_{j=1}^{a}\left(1-\frac{\left|(s_i+1)-s_j^l\right|}{r_s}-\widehat{Sim}\right)+\sum_{j=1}^{b}\left(1-\frac{\left|(s_i+1)-s_j^e\right|}{r_s}-\widehat{Sim}\right)\right.$$

$$\left.+\sum_{j=1}^{c}\left(1-\frac{\left|(s_i+1)-s_j^h\right|}{r_s}-\widehat{Sim}\right)\right)$$

$$=\frac{1}{a+b+c}\left(a+b+c-\frac{as_i+a-\sum_{j=1}^{a}s_l^j-b+\sum_{j=1}^{c}s_h^j-cs_i-c}{r_s}\right)-\widehat{Sim}$$

Thus the average similarity score would change by

$$\frac{1}{a+b+c}\left(a+b+c-\frac{(a-c)s_i-\sum_{j=1}^{a}s_j^l+\sum_{j=1}^{c}s_j^h+a-b-c}{r_s}\right)-\widehat{Sim}$$

$$-\left(\frac{1}{a+b+c}\left(a+\mathrm{b}+c-\frac{(a-c)s_i-\sum_{j=1}^{a}s_j^l+\sum_{j=1}^{c}s_j^h}{r_s}\right)-\widehat{Sim}\right)$$

$$=\frac{c-a-b}{(a+b+c)r_s}$$

Therefore, the contribution of the average similarity effect to the log-probability ratio of increasing this ego's stress level by one unit would be $\beta_{avSim}\frac{c-a-b}{(a+b+c)r_s}$. These expressions make clear that the influence of social contacts on stress change depends only on the proportion of those whose stress levels are higher than that of ego, not on how much larger or smaller the values are. For example, in our study, the estimate of the average similarity effect is 6.65. For an individual who has five friends, among whom three score higher and two score lower, the contribution of the average similarity effect to the odds for this person moving up instead of remaining unchanged would be $exp\left(6.65*\frac{3-2}{(3+2)*4}\right)=1.39$.

The interaction effect

The interaction between average similarity and peer variance is defined as

$$avSimAltVar_i(x,s) = avSim_{ij}(x,s) * altVar_i(x,s)$$

$$= \left(\frac{1}{\sum_j x_{ij}} \sum_j x_{ij}(Sim_{ij} - \widehat{Sim})\right)$$

$$* \left(\frac{1}{\sum_j x_{ij} - 1} \sum_j x_{ij}\,(s_j - \bar{s}_{alt(i)})^2 - Var^{obs}\right)$$

Since the purpose of our study is to understand how social influence varies as a function of peer variance, we want to show how the value of the average similarity effect changes at each specific peer variance level. For a person *i* with a specific peer variance value $altVar_i$, the moderated social influence from social contacts is $\beta_{avSim} + \beta_{avSimAltVar} * (altVar_i - Var^{obs})$. Thus the contribution of the average similarity effect to the odds of increasing stress compared with not to increase is

$$exp\left(\left(\beta_{avSim} + \beta_{avSimAltVar} * (altVar_i - Var^{obs})\right)\frac{c-a-b}{(a+b+c)r_s}\right)$$

This expression is plotted in Figure 1 in the main manuscript, for the parameters estimated in this study. The x-axis represents the proportion of friends whose stress is higher than ego's stress, and the y-axis represents the contribution of average similarity effect to the odds of increasing stress compared again no change. Different lines in the figures indicate different levels of peer variance.

The peer variance effect (varAlt) and the interaction effect between average similarity and peer variance (avSimVarAlt) have been implemented in the RSiena (Simulation Investigation for Empirical Network Analysis; Ripley et al., 2022) statistical package in R for the purpose of this study.

The real odds

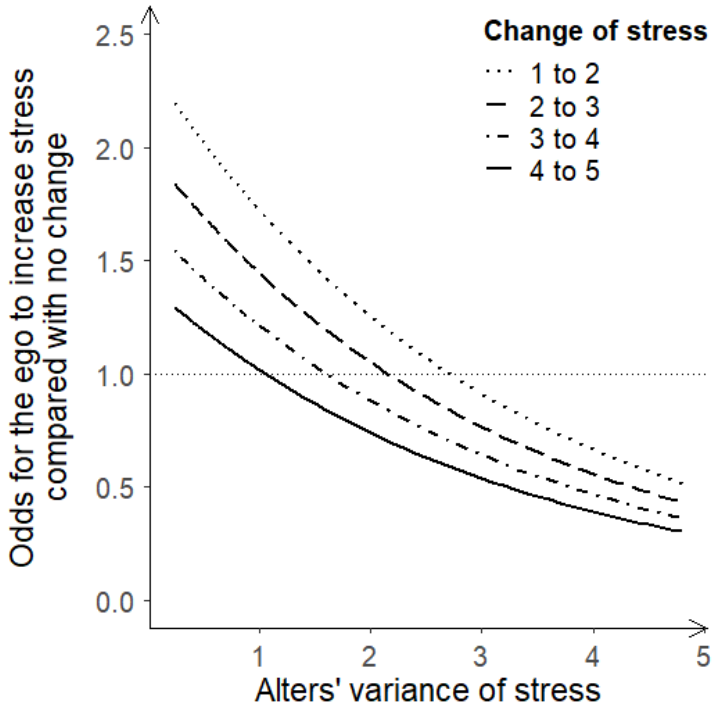
Note that Figure 1 shows the contribution of the average similarity effect to the odds for various values of the peer variance. The actual odds to increase stress by one level also depend on the other effects that are included in the stress objective function. To obtain these odds, we also need to consider the contribution of the linear and quadratic shape effects (Snijders et al., 2007). Suppose we have an actor i, whose continuous attributes, such as working load and neuroticism score, are all at the average level, and categorical attributes, such as gender, are at reference level (i.e., coded as 0) in our analysis (which means actor a is a male student).

Since continuous covariates are centered, the average level equals to 0. And the reference level of categorical covariates also equals to 0. Then if this actor i does not change any of these covariates, the contribution of these variables to the objective function will always remain 0. Therefore, for this actor i, the odds of increasing stress from the current level s_i^c to one unit higher $s_i^c + 1$ compared with no change are given by

$$
\begin{aligned}
&exp\big(f_i(x, s') - f_i(x, s)\big) \\
&\quad = exp\Bigg(\Big(\beta_{linear} * \big((s_i^c - \hat{s} + 1) - s_i^c - \hat{s}\big)\Big) \\
&\quad + \Big(\beta_{quadratic} * ((s_i^c - \hat{s} + 1)^2 - (s_i^c - \hat{s})^2)\Big) \\
&\quad + \Big(\beta_{avSim} + \beta_{avSimAltVar} * (altVar - Var^{obs})\Big)\frac{c - a - b}{(a + b + c)r_s}\Bigg) \\
&\quad = exp\Big(\beta_{linear} + \beta_{quadratic} * (2(s_i^c - \hat{s}) + 1) \\
&\quad + \Big(\beta_{avSim} + \beta_{avSimAltVar} * (altVar - Var^{obs})\Big)\frac{c - a - b}{(a + b + c)r_s}\Big)
\end{aligned}
$$

where $\hat{s}$ denotes the mean of stress over the four measurement points. In Figure 3, we show how the odds change with the peer variance, for an individual, three-fifths of whose friends score higher in stress. For example, if this person has five friends, then three of them score higher and two score lower or equal (e.g., $a = 2$, $b = 0$, $c = 3$). The x-axis here represents the peer variance, while the y-axis represents the odds of increasing stress compared with no change. Each line indicates a specific change from one level to one unit higher. We should notice that since we assume 60% of the alters have higher stress than the ego, the range of peer variance that can be achieved is between 0.24 and 4.8. For example, consider an "average" actor whose original stress level is 4. Three-fifths of their alters should have stress equal to 5, and two-fifths of their alters should have stress equal to or lower than 4. The highest peer variance of stress can be achieved when those in the latter group all have a stress score of 1. And the lowest peer variance of stress can be achieved when they have a stress level of 4. Congruent with Figure 1, the odds of increasing stress by one level compared with no change are high when peers have consensus in stress perceptions (low peer variance). As the discrepancy among peers climbs (high peer variance), the ego becomes less likely to adjust their stress to resemble their peers' level. Figure 3 shows that this pattern holds regardless of the individual's initial level of stress. It is also worth noticing that the odds for an individual to increase their stress compared with no change is higher when this person's initial stress level is low. For example, when the peer variance is 0.3, the odds of increasing stress from 1 to 2 compared with no change are 2.15, while the odds of increasing stress from 4 to 5 compared with no change are 1.27. When the peer variance is 1.2, the corresponding odds for these two scenarios are 1.62 and 0.95.

Figure S1. The odds for individuals to increase stress compared with no change at different levels of alters' variance of stress when three-fifths of their friends score higher than them



Supplemental 2. Interpreting the covariate similarity effect in the network dynamics model

The similarity parameter can be interpreted as a log odds ratio. Specifically, assume an actor i, whose stress level is 1, currently has no tie to either j or h, and that j and h have equivalent network positions and the same values on all covariates, except for scoring 1 and 5 on the stress scale, respectively. Then, the similarity score of stress between i and j is 1 while that between i and h is 0 (see Supplemental 1 for the definition of the similarity score). In this situation, the probability ratio for i to befriend j compared to h is $e^{0.32} = 1.38$.

Supplemental 3

Table S1. Description and interpretation of RSiena effects

Effects	RSiena term	Description of positive effect
Behavioral Dynamics		
Social influence-related effects		
Average stress similarity	avSim	Tendency of individuals' stress to becoming similar to their outgoing social contacts' stress
Average stress similarity x Peer stress variance	avSimVarAlt	Tendency of individuals' stress to becoming more similar to their outgoing social contacts' stress when the variance of these social contacts' stress is higher
Average stress similarity x Neuroticism ego	avSimEgoX	Tendency of individuals with higher neuroticism to becoming more similar to their outgoing social contacts with respect to stress
Average stress similarity x Conscientiousness ego	avSimEgoX	Tendency of individuals with higher conscientiousness to becoming more similar to their outgoing social contacts with respect to stress
Average stress similarity x Locus of control ego	avSimEgoX	Tendency of individuals with higher locus of control to becoming more similar to their outgoing social contacts with respect to stress
Individual covariates on stress		
Peer variance	varAlt	Tendency for individuals whose social contacts have a higher variance of stress to develop higher stress
Neuroticism	effFrom	Tendency for individuals with higher neuroticism to develop higher stress
Conscientiousness	effFrom	Tendency for individuals with higher conscientiousness to develop higher stress
Locus of Control	effFrom	Tendency for individuals with higher locus of control to develop higher stress
Outdegree	outdeg	Tendency for individuals who send more outgoing expressive ties to develop higher stress
Female	effFrom	Tendency for females (than males) to develop higher stress
US	effFrom	Tendency for US students (than those from other countries) to develop higher stress
Asian	effFrom	Tendency for Asian students (than those of other ethnicities) to develop higher stress
Workload	effFrom	Tendency for individuals with higher workload to develop higher stress
Outside support	effFrom	Tendency for individuals with more outside support to develop higher stress
Rate function		
Rate period (stress)		Frequency of stress changes between each two consecutive time points

Basic effects

Linear shape	linear	Basic drive toward high values on stress
Quadratic shape	quad	A positive quadratic shape effect indicates a self-reinforcing tendency in stress development (i.e., more extreme stress scores become even more extreme); a negative quadratic shape effect indicates a self-correcting tendency in stress development (i.e., more extreme stress scores are more strongly regressed toward the middle value)

Network Dynamics

Rate function

Rate period (expressive ties)		Frequency of expressive tie changes between each two consecutive time points

Structural effects

Outdegree	density	Tendency to send outgoing expressive ties
Reciprocity	recip	Tendency to reciprocate expressive ties
GWESP I -> K -> J	gwespFF	Tendency to send expressive ties to outgoing social contacts' outgoing social contacts, a proxy of transitivity
GWESP I <- K <- J	gwespBB	Tendency to send expressive ties to incoming social contacts' incoming social contacts, a proxy of cyclicity
Indegree popularity (sqrt)	inPopSqrt	Tendency for those receiving many incoming expressive ties to receive even more incoming expressive ties
Outdegree popularity (sqrt)	outPopSqrt	Tendency for those sending many outgoing expressive ties to receive even more incoming expressive ties
Outdegree activity (sqrt)	outActSqrt	Tendency for those sending many outgoing expressive ties to send even more outgoing expressive ties
Reciprocity x GWESP I -> K -> J	recip x gwespFF	Tendency to reciprocate expressive ties to outgoing social contacts' outgoing social contacts

Stress-related selection effects

Stress ego	egoX	Tendency for those with higher stress to send more outgoing expressive ties
Stress alter	alterX	Tendency for those with higher stress to receive more incoming expressive ties
Stress similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar stress to themselves

Individual covariates on network

Female ego	egoX	Tendency for females (than males) to send more outgoing expressive ties
Female alter	alterX	Tendency for females (than males) to receive more incoming expressive ties
Same gender	X	Tendency for individuals to send outgoing expressive ties to those of the same gender

Same ethnicity	X	Tendency for individuals to send outgoing expressive ties to those of the same ethnicity
Same citizenship	X	Tendency for individuals to send outgoing expressive ties to those of the same citizenship
Same program	X	Tendency for individuals to send outgoing expressive ties to those enrolled in the same academic program
Same orientation squad	X	Tendency for individuals to send outgoing expressive ties to those who are assigned to the same oriantation squad during the orientation week
Same course section	X	Tendency for individuals to send outgoing expressive ties to those who attend more same course sections as themselves
Neuroticism similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar neuroticism to themselves
Conscientiousness similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar conscientiousnes to themselves
Extraversion similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar extraversion to themselves
Agreeableness similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar agreeableness to themselves
Openness similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar openness to themselves
Locus of Control similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar locus of control to themselves
Workload similarity	simX	Tendency for individuals to send outgoing expressive ties to those with similar workload to themselves

Supplemental 4

Table S2. The breakdown of participants' demographic and situational characteristics

	All	T1	T2	T3	T4
Num. of respondents	321	309	276	256	241
Ethnicity					
White	94 (29.3%)	93 (30.1%)	86 (31.1%)	82 (32.0%)	78 (32.4%)
Asian	174 (54.2%)	168 (54.4%)	146 (52.9%)	134 (52.3%)	126 (52.3%)
Black	12 (3.7%)	11 (3.6%)	11 (4.0%)	10 (3.9%)	10 (4.1%)
Hispanic	7 (2.2%)	7 (2.3%)	6 (2.2%)	6 (2.4%)	5 (2.1%)
Multi	11 (3.4%)	10 (3.2%)	8 (2.9%)	6 (2.4%)	6 (2.5%)
Other	23 (7.2%)	20 (6.5%)	19 (6.9%)	18 (7.0%)	16 (6.6%)
Citizenship					
US	129 (40.2%)	126 (40.8%)	114 (41.3%)	107 (41.8%)	104 (43.2%)
China	66 (20.6%)	63 (20.4%)	52 (18.8%)	46 (18.0%)	42 (17.4%)
India	96 (29.9%)	93 (30.1%)	86 (31.2%)	83 (32.4%)	78 (32.4%)
Other American	10 (3.1%)	9 (2.9%)	9 (3.3%)	7 (2.7%)	7 (2.9%)
Other	20 (6.2%)	18 (5.8%)	15 (5.4%)	13 (5.1%)	10 (4.1%)
Gender					
Female	174 (54.2%)	168 (54.4%)	151 (54.7%)	140 (54.7%)	132 (54.8%)
Male	142 (44.2%)	136 (44.0%)	121 (43.8%)	112 (43.8%)	106 (44.0%)
Other	5 (1.6%)	5 (1.6%)	4 (1.5%)	4 (1.5%)	3 (1.2%)
Academic program					
MISM	159 (49.5%)	153 (49.5%)	140 (50.7%)	133 (52.0%)	123 (51.0%)
MSPPM	89 (27.7%)	86 (27.8%)	75 (27.2%)	67 (26.2%)	65 (27.0%)
MSHC	18 (5.6%)	17 (5.5%)	11 (4.0%)	11 (4.3%)	10 (4.1%)
MAM	20 (6.2%)	19 (6.1%)	18 (6.5%)	17 (6.6%)	17 (7.1%)
Other	35 (10.9%)	34 (11.0%)	32 (11.6%)	28 (10.9%)	26 (10.8%)

Note: The demographic information was provided by Student Affairs Office. Ethnicity, citizenship, and gender were self-reported by the student. We aggregated the citizenship by regarding students from American countries other than the US as Other American (e.g., Canada, Mexico, and Colombia), and countries other than US, China, India, and other American countries as Other (e.g., Australia, Germany, and Korea). The graduate school has twelve academic programs. We incorporated sub-tracks of a program into one program. And we aggregated the programs with fewer than ten students into the category Other here.

Supplemental 5

Table S3. The correlation matrix for key variables

		Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12
1	Ego stress T1	3.15	1.03	1											
2	Ego stress T2	3.26	1.08	0.46	1										
3	Ego stress T3	3.42	1.07	0.30	0.61	1									
4	Ego stress T4	3.57	1.18	0.25	0.55	0.70	1								
5	Average alters' stress T1	3.12	0.51	0.13	0.44	0.27	0.23	1							
6	Average alters' stress T2	3.24	0.63	0.24	0.29	0.48	0.34	0.40	1						
7	Average alters' stress T3	3.44	0.59	0.23	0.40	0.43	0.53	0.49	0.64	1					
8	Average alters' stress T4	3.56	0.61	0.21	0.35	0.41	0.44	0.42	0.57	0.71	1				
9	Alters' variance of stress T1	1.05	0.65	0.00	-0.04	-0.04	-0.05	-0.08	-0.05	-0.06	-0.10	1			
10	Alters' variance of stress T2	1.03	0.91	0.00	-0.11	-0.10	-0.13	-0.04	-0.27	-0.28	-0.28	-0.10	1		
11	Alters' variance of stress T3	0.93	0.76	-0.06	-0.16	-0.22	-0.26	-0.31	-0.38	-0.48	-0.45	-0.02	0.25	1	
12	Alters' variance of stress T4	1.16	0.84	-0.14	-0.25	-0.22	-0.29	-0.34	-0.39	-0.50	-0.60	0.03	0.33	0.39	1

Supplemental 6

Table S4. Results of the SAOM stress and the friendship co-evolution (full)

	Model 1		Model 2		Model 3		Model 4	
	Parameter Estimates	**Confidence Interval (95%)**	**Parameter Estimates**	**Confidence Interval (95%)**	**Parameter Estimates**	**Confidence Interval (95%)**	**Parameter Estimates**	**Confidence Interval (95%)**
Behavior dynamics								
Average similarity	7.55 *** (1.01)	[5.57, 9.53]	10.24 *** (1.79)	[6.73, 13.75]	6.65 *** (1.65)	[3.42, 9.88]	6.66 *** (1.64)	[3.45, 9.87]
Average similarity * peer variance			-5.39 ** (1.77)	[-8.86, -1.92]	-6.35 *** (1.69)	[-9.66, -3.04]	-6.57 *** (1.74)	[-9.98, -3.16]
Average similarity * neuroticism			3.65 ** (1.24)	[1.22, 6.08]	3.18 ** (1.11)	[1.00, 5.36]	3.21 ** (1.08)	[1.09, 5.33]
Average similarity * conscientiousness			-5.32 *** (1.55)	[-8.36, -2.28]	-4.92 *** (1.38)	[-7.43, -2.41]	-4.94 *** (1.38)	[-7.64, -2.24]
Average similarity * locus of control			-4.62 * (2.08)	[-8.70, -0.54]	-3.63 * (1.80)	[-7.16, -0.10]	-3.65 * (1.72)	[-7.02, -0.28]
Network dynamics								
Stress alter	-0.036 * (0.018)	[-0.071, -0.00072]	-0.052 * (0.021)	[-0.093, -0.011]	-0.048 (0.021)	[-0.089, -0.0068]	-0.051 * (0.021)	[-0.092, -0.0098]
Stress ego	-0.045 * (0.021)	[-0.086, -0.0038]	-0.063 * (0.025)	[-0.11, -0.014]	-0.078 *** (0.024)	[-0.13, -0.031]	-0.079 *** (0.023)	[-0.12, -0.034]
Stress similarity	0.48 *** (0.14)	[0.21, 0.75]	0.55 ** (0.17)	[0.22, 0.88]	0.32 * (0.16)	[0.0064, 0.63]	0.34 * (0.16)	[0.026, 0.65]
Covariates for behavior dynamics								
Peer variance			-0.26 (0.26)	[-0.77, 0.25]	-0.074 (0.32)	[-0.70, 0.55]	-0.099 (0.34)	[-0.77, 0.57]
Gender (female = 1)					0.0075 (0.098)	[-0.18, 0.20]		
Ethnicity (White = 1)					-0.041 (0.14)	[-0.32, 0.23]		
Citizenship (US = 1)					-0.40 ** (0.15)	[-0.69, -0.11]	-0.40 ** (0.14)	[-0.67, -0.13]
Neuroticism			0.24 *** (0.062)	[0.12, 0.36]	0.26 *** (0.061)	[0.14, 0.38]	0.25 *** (0.058)	[0.14, 0.36]
Conscientiousness			-0.19 * (0.082)	[-0.35, -0.029]	-0.22 ** (0.080)	[-0.38, -0.063]	-0.22 ** (0.081)	[-0.38, -0.061]
Locus of control			-0.23 * (0.10)	[-0.43, -0.034]	-0.25 * (0.11)	[-0.47, -0.034]	-0.23 * (0.11)	[-0.45, 0.014]
Extraversion					-0.095 (0.052)	[-0.02, 0.0069]		
Agreeableness					0.16 (0.091)	[-0.018, 0.34]		
Openness					0.024 (0.074)	[-0.12, 0.17]		
Working load (hours)					0.027 *** (0.0067)	[0.014, 0.04]	0.026 *** (0.0066)	[0.013, 0.039]
Outdegree					0.024 ** (0.0090)	[0.0064, 0.042]	0.025 * (0.0099)	[0.0056, 0.044]
Outside support					-0.022 * (0.010)	[-0.042, -0.0024]	-0.020 (0.013)	[-0.045, 0.0055]
Covariates for network dynamics								
Outdegree (density)	-2.31 *** (0.064)	[-2.44, -2.18]	-2.33 *** (0.066)	[-2.46, -2.20]	-2.61 *** (0.068)	[-2.74, -2.48]	-2.62 *** (0.068)	[-2.75, -2.49]
Reciprocity	2.66 *** (0.071)	[2.52, 2.80]	2.65 *** (0.071)	[2.51, 2.79]	2.35 *** (0.072)	[2.21, 2.49]	2.35 *** (0.071)	[2.21, 2.49]

GWESP I -> K -> J	1.54 *** (0.041)	[1.46, 1.62]	1.54 *** (0.041)	[1.46, 1.62]	1.30 *** (0.042)	[1.22, 1.38]	1.30 *** (0.043)	[1.22, 1.38]
GWESP I <- K <- J	0.31 *** (0.043)	[0.23, 0.39]	0.31 *** (0.045)	[0.22, 0.40]	0.18 *** (0.046)	[0.090, 0.27]	0.18 *** (0.048)	[0.086, 0.27]
Indegree popularity (sqrt)	0.048 (0.026)	[-0.003, 0.099]	0.048 (0.026)	[-0.0030, 0.099]	0.086 *** (0.026)	[0.035, 0.14]	0.086 ** (0.027)	[0.033, 0.14]
Outdegree popularity (sqrt)	-0.37 *** (0.031)	[-0.43, -0.31]	-0.37 *** (0.032)	[-0.43, -0.31]	-0.37 *** (0.032)	[-0.43, -0.31]	-0.37 *** (0.034)	[-0.44, -0.30]
Outdegree activity (sqrt)	-0.0026 (0.0097)	[-0.022, 0.016]	-0.00094 (0.0097)	[-0.020, 0.018]	0.033 *** (0.0098)	[0.014, 0.052]	0.033 *** (0.0097)	[0.014, 0.052]
Reciprocity x GWESP I -> K -> J	-0.89 *** (0.059)	[-1.01, -0.77]	-0.89 *** (0.057)	[-1.00, -0.78]	-0.74 *** (0.058)	[-0.85, -0.63]	-0.75 *** (0.058)	[-0.86, -0.64]
Same academic program					0.29 *** (0.028)	[0.24, 0.34]	0.29 *** (0.027)	[0.24, 0.34]
Same orientation squad					0.54 *** (0.044)	[0.45, 0.63]	0.53 *** (0.044)	[0.44, 0.62]
Same course					0.21 *** (0.013)	[0.18, 0.24]	0.21 *** (0.013)	[0.18, 0.24]
Working load (hours) similarity					0.47 *** (0.14)	[0.20, 0.74]	0.47 *** (0.14)	[0.20, 0.74]
Same ethnicity					0.066 * (0.027)	[0.013, 0.12]	0.066 * (0.027)	[0.013, 0.12]
Same citizenship					0.42 *** (0.028)	[0.37, 0.47]	0.41 *** (0.029)	[0.35, 0.47]
Same gender					0.16 *** (0.023)	[0.11, 0.21]	0.16 *** (0.023)	[0.11, 0.21]
Gender alter (female = 1)					-0.029 (0.023)	[-0.074, 0.016]		
Gender ego (female = 1)					-0.061 * (0.027)	[-0.11, -0.0081]	-0.071 * (0.029)	[-0.13, -0.014]
Neuroticism similarity					-0.069 (0.081)	[-0.23, 0.090]		
Conscientiousness similarity					0.16 (0.084)	[-0.0046, 0.32]		
Locus of control similarity					-0.038 (0.079)	[-0.19, 0.12]		
Extraversion similarity					0.22 ** (0.080)	[0.063, 0.38]	0.24 ** (0.079)	[0.085, 0.39]
Agreeableness similarity					-0.015 (0.083)	[-0.18, 0.15]		
Openness similarity					0.16 (0.085)	[-0.0066, 0.33]		
Basic model effects								
Behavior rate (period 1)	3.77 *** (0.58)	[2.63, 4.91]	4.44 *** (0.90)	[2.68, 6.20]	4.82 *** (1.02)	[2.82, 6.82]	4.69 *** (1.11)	[2.51, 6.87]
Behavior rate (period 2)	2.87 *** (0.41)	[2.07, 3.67]	6.96 *** (1.98)	[3.08, 10.84]	7.33 ** (2.54)	[2.35, 12.31]	6.92 ** (2.66)	[1.71, 12.13]
Behavior rate (period 3)	2.52 *** (0.38)	[1.78, 3.26]	5.37 *** (1.59)	[2.25, 8.49]	4.59 *** (1.07)	[2.49, 6.69]	4.46 *** (1.23)	[2.05, 6.87]
Behavior linear shape	0.21 *** (0.043)	[0.13, 0.29]	0.26 *** (0.071)	[0.12, 0.40]	0.036 (0.13)	[-0.22, 0.29]	0.023 (0.14)	[-0.25, 0.30]
Behavior quadratic shape	0.27 *** (0.053)	[0.17, 0.37]	0.13 (0.069)	[-0.0052, 0.27]	-0.088 (0.09)	[-0.26, 0.088]	-0.085 (0.097)	[-0.28, 0.11]
Network rate (period 1)	21.85 *** (0.91)	[20.07, 23.63]	21.63 *** (0.97)	[19.73, 23.53]	23.11 *** (1.07)	[21.01, 25.21]	23.13 *** (1.11)	[20.95, 25.31]
Network rate (period 2)	19.56 *** (0.72)	[18.15, 20.97]	19.58 *** (0.73)	[18.15, 21.01]	20.65 *** (0.77)	[19.14, 22.16]	20.65 *** (0.79)	[19.10, 22.20]
Network rate (period 3)	14.57 *** (0.52)	[13.55, 15.59]	14.56 *** (0.52)	[13.54, 15.58]	14.92 *** (0.52)	[13.90, 15.94]	14.93 *** (0.51)	[13.93, 15.93]

Note. N = 309. * p < .05 ** p < .01 *** p < .001 (two-tailed tests). All models converged successfully. Overall maximum convergence ratio < 0.25; all t-statistics for deviations < 0.10 in absolute value for each parameter estimate. Reciprocity indicates the tendency to reciprocate incoming ties. GWESP I -> K -> J reflects the tendency to create ties with "friends of friends," and GWESP I <- K <- J reflects the tendency to form cyclic triadic structures. Outdegree popularity indicates the tendency to send ties to actors who send many ties. Indegree popularity indicates the tendency to send ties to actors who receive many ties.

Outdegree activity indicates the tendency of actors with many outgoing ties to create more ties. Reciprocity × GWESP I -> K -> J indicates the tendency to reciprocate ties with “friends-of-friends.”

Supplemental 7

Table S5. Comparison of T1 characteristics of dropped-out participants for each wave

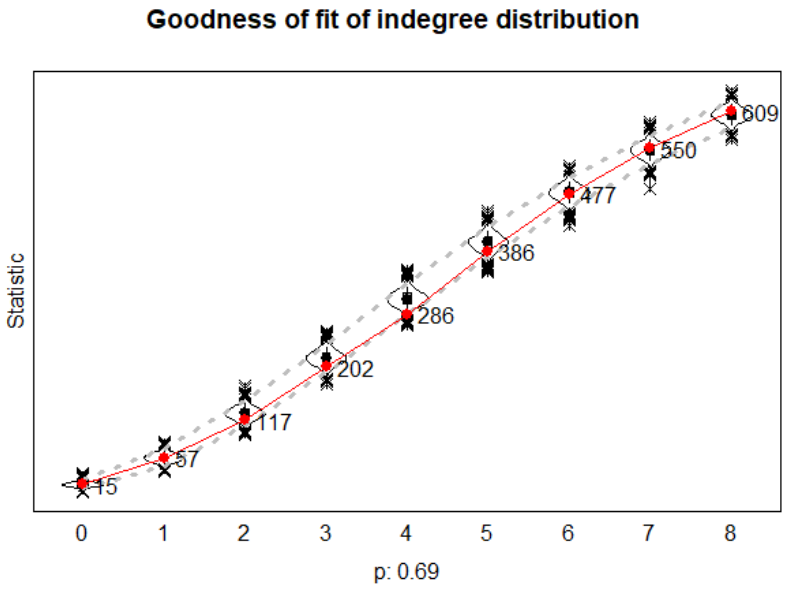
	Not participate	Drop out after T1	Drop out after T2	Drop out after T3	Fully participate
Number	12	33	20	15	241
Gender (% Female)	6 (50.0%)	17 (51.5%)	11 (55.0%)	8 (53.3%)	132 (54.8%)
Ethnicity (% Asian)	6 (50.0%)	22 (66.7%)	12 (60.0%)	8 (53.3%)	78 (32.4%)
Citizenship (% US)	3 (25.0%)	12 (36.4%)	7 (35.0%)	3 (20.0%)	104 (43.2%)
Program (% MISM)	6 (50.0%)	13 (39.4%)	7 (35.0%)	10 (66.7%)	123 (51.0%)
Perceived stress (T1)		3.12 (0.99)	2.90 (0.85)	3.27 (1.16)	3.16 (1.05)
Working load (T1)		41.97 (8.03)	41.01 (9.42)	45.66 (6.05)	43.50 (7.75)
Outside support (T1)		3.52 (4.18)	2.45 (2.76)	2.73 (2.94)	3.29 (3.48)
Neuroticism		3.77 (1.02)	3.79 (0.89)	3.40 (1.36)	3.53 (1.00)
Conscientiousness		4.66 (0.89)	4.69 (0.83)	4.54 (1.18)	4.74 (0.92)
Locus of control		4.01 (0.16)	3.91 (0.59)	3.79 (0.75)	3.91 (0.53)
Agreeableness		5.69 (0.59)	5.41 (0.65)	5.36 (0.59)	5.40 (0.64)
Extraversion		4.25 (1.15)	4.19 (1.17)	4.71 (1.48)	4.42 (1.01)
Openness		5.51 (0.73)	5.37 (0.83)	5.20 (0.82)	5.39 (0.76)
Outdegree (T1)		6.45 (4.48)	4.95 (2.16)	6.53 (3.60)	6.61 (3.94)
Indegree (T1)		5.58 (3.27)	4.40 (3.30)	7.20 (3.70)	6.54 (4.03)

Note. This table shows the comparison between five subgroups: those who chose not to participate in the study at the very beginning (Not participate), those who participated in the first wave yet dropped in the second wave (Drop out after T1), those who participated in the first two waves yet dropped in the third wave (Drop out after T2), those who participated in the first three waves yet dropped in the last wave (Drop out after T3), and those who participated for all four waves (Fully participate). We compute the mean and standard deviation of time-variant variables for these three subgroups based on the first-wave data. Outdegree and indegree indicate the number of friendship (or advice) nominations made and received, respectively. Individual-related features only are available for those who at least participated at Time 1 (and signed the consent form). As shown, the demographic compositions, situational, individual, and network features do not differ significantly between the five subgroups.

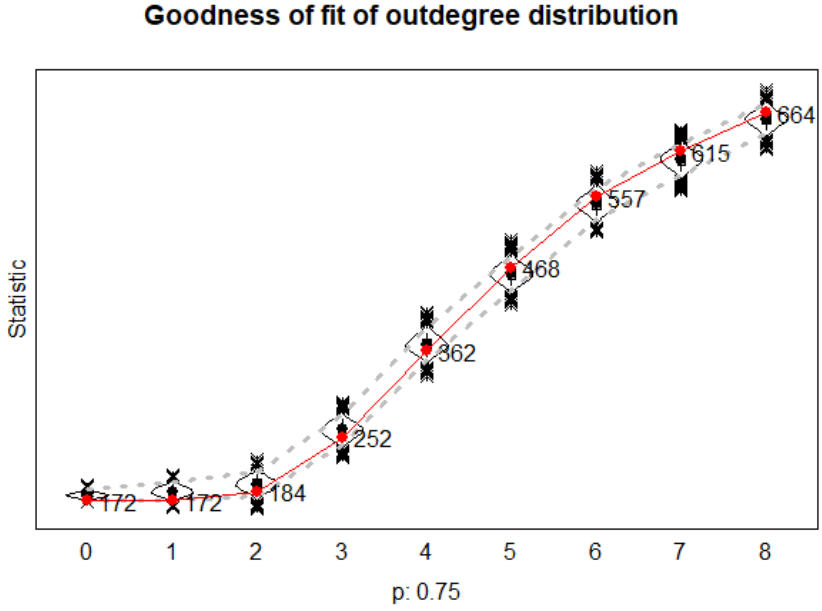
Supplemental 8. Goodness of fit for the friendship network and perceived stress

After estimating the models, we ran Mahalanobis distance-based Monte Carlo goodness of fit (GOF) tests to determine how well the model reproduced features of the observed data not explicitly fit by the models (Lospinoso & Snijders, 2019). In particular, we compared four important features of the observed network and participants' stress to their expected values in the estimated distribution of a large number of simulated networks. These four features were: 1) outdegree distribution (frequencies of outgoing ties); 2) indegree distribution (frequencies of incoming ties); 3) triad census (frequencies of 16 possible isomorphic subgraphs of three nodes in the network; Borgatti, Everett, & Johnson, 2018; Holland & Leinhardt, 1976); 4) geodesic distances (frequencies of shortest path length) (Hunter, Goodreau, & Handcock, 2008; Lospinoso & Snijders, 2019). We also examined whether the change in the behavioral variable (i.e., stress) was appropriately fit by the model. Based on 10000 simulations, we found that the model fit was adequate for the indegree, outdegree, geodesic distance, and change in stress between consecutive waves.

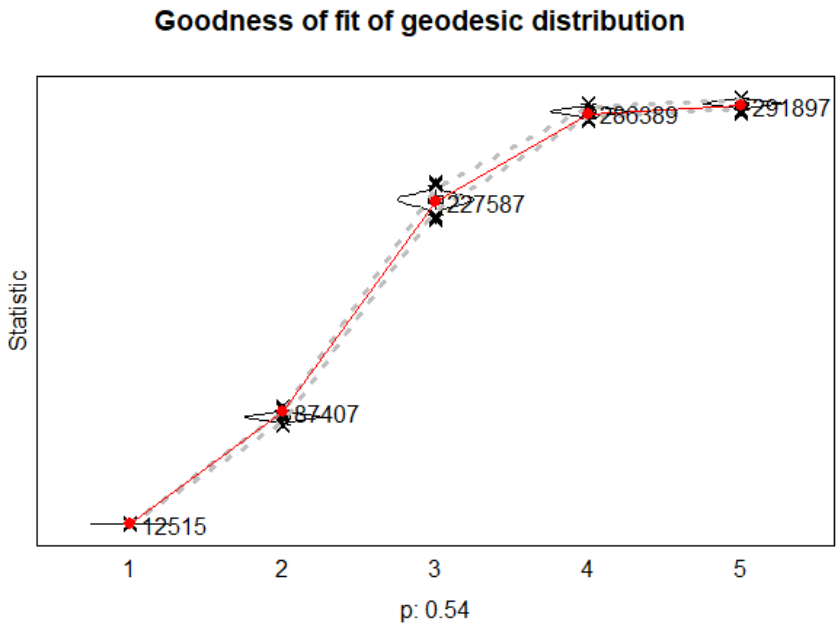
The counts of two subgraphs (specifically, 120U, 300) in the triad census (Borgatti et al., 2018; Holland & Leinhardt, 1976) were modeled inadequately. We estimated several additional models trying to improve the model fit for these statistics, including more structural effects. Some of these model modifications improved the model fit for the three subgraph counts, yet worsened the fit for others. Therefore, we retained the model specification as presented in this section. Note that all of the more elaborate models we estimated corroborated the main findings of our study.



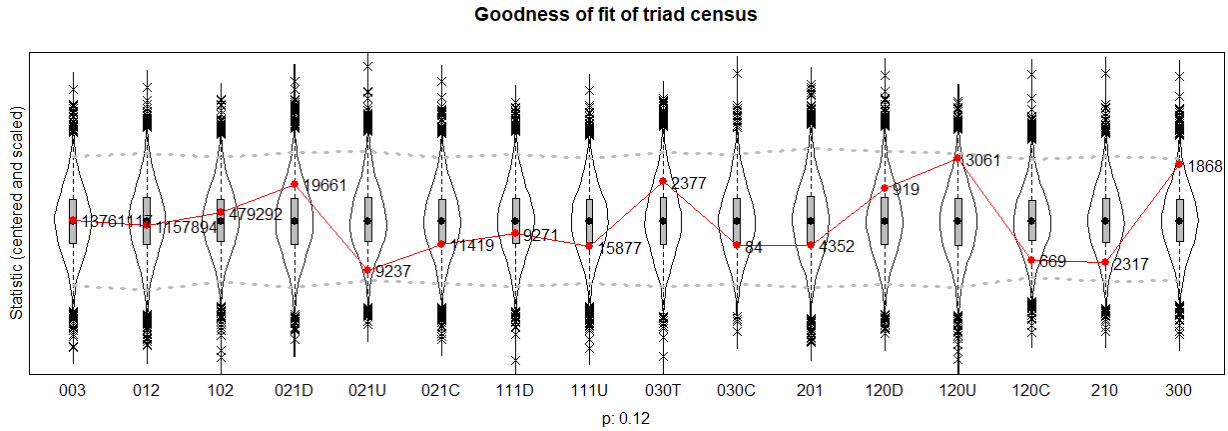
Note: The x-axis represents the number of incoming ties. The numbers in the plot connected by the solid line indicate the observed *cumulative* indegree distribution aggregated over the last three waves. The violin plots represent simulated statistics. Dotted lines give 95th percentile bands. The p-value for the Mahalanobis distance-based Monte Carlo goodness of fit test is given at the bottom.



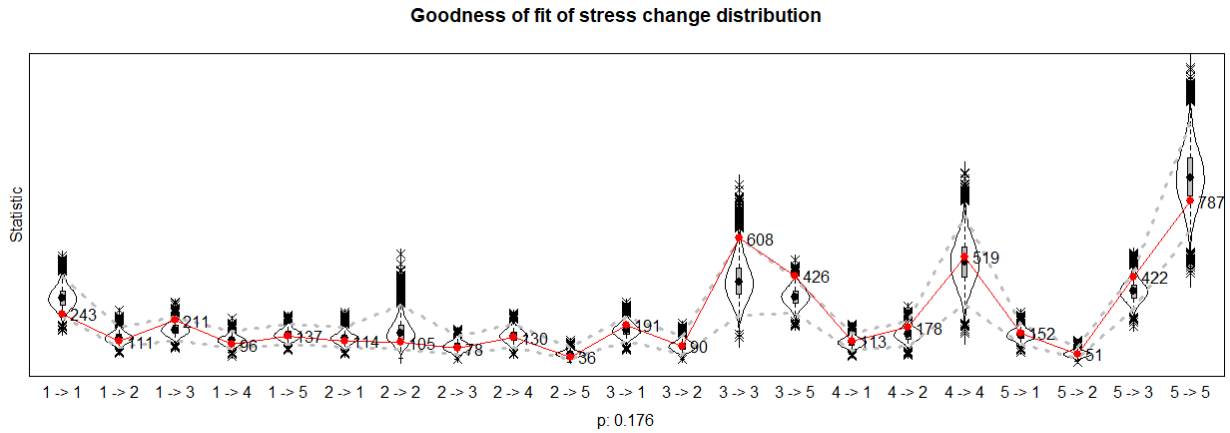
Note: The x-axis represents the number of outgoing ties. The numbers in the plot connected by the solid line indicate the *cumulative* observed outdegree distribution aggregated over the last three waves. The violin plots represent simulated statistics. Dotted lines give 95th percentile bands. The p-value for the Mahalanobis distance-based Monte Carlo goodness of fit test is given at the bottom.



Note: The x-axis represents geodesic distance, as the length of the shortest path between nodes *i* and *j* in the graph, without taking into account the directionality of edges. The numbers in the plot connected by the solid line indicate the observed *cumulative* geodesic distribution aggregated over the last three waves. The violin plots represent simulated statistics. Dotted lines give 95th percentile bands. The p-value for the Mahalanobis distance-based Monte Carlo goodness of fit test is given at the bottom.



Note: The x-axis represents the potential outcomes in the triad census: the collection of the 16 possible isomorphic subgraphs of three nodes (Borgatti et al., 2018; Holland & Leinhardt, 1976). The numbers in the plot connected by the solid line indicate the observed triad census distribution (observations are centered and scaled) aggregated over the last three waves. The violin plots represent simulated statistics. Dotted lines give 95th percentile bands. The p-value for the Mahalanobis distance-based Monte Carlo goodness of fit test is given at the bottom.



Note: The x-axis represents the 25 possible changes in stress levels between consecutive measurement moments. The numbers in the plot connected by the solid line indicate the observed stress change distribution aggregated over the last three waves. The violin plots represent simulated statistics. Dotted lines give 95th percentile bands. The p-value for the Mahalanobis distance-based Monte Carlo goodness of fit test is given at the bottom.

Supplemental 9

Table S6. Specification of the simulated Stochastic Actor-oriented Models in the effect-size sensitivity analysis

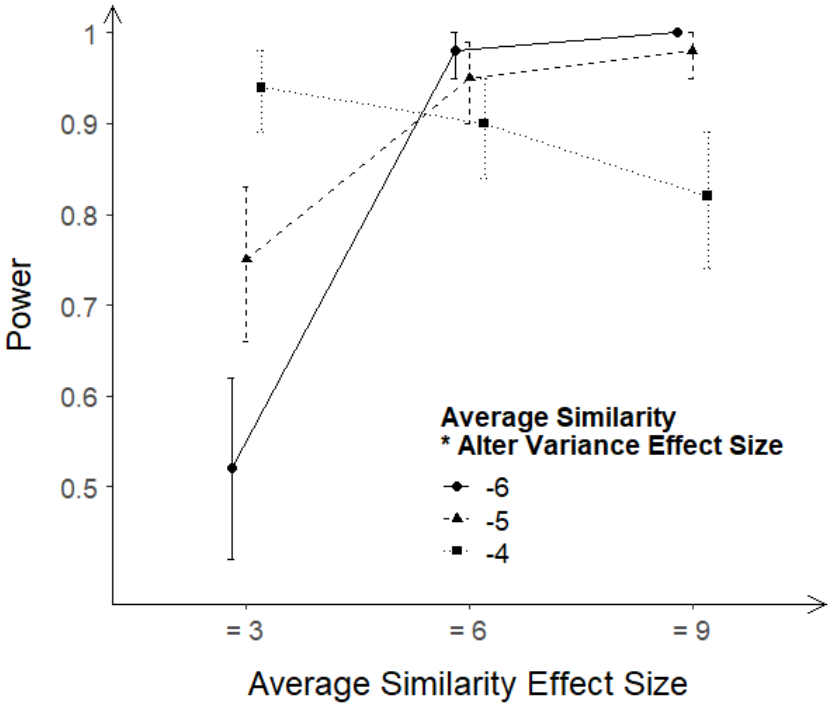
Effect	RSiena term	Parameter
Stress Dynamics		
Average similarity	avSim	in {3, 6, 9}
Average similarity * peer variance	avSimVarAlt	in {-4, -5, -6}
Peer variance	varAlt	-0.20
Change between t1 and t2	rate 1	4.50
Change between t2 and t3	rate 2	7.00
Change between t3 and t4	rate 3	5.00
Linear shape	linear	0.20
Quadratic shape	quad	0.10
Network Dynamics		
Outdegree	density	-2.30
Reciprocity	recip	2.65
GWESP I -> K -> J	gwespFF	1.54
GWESP I <- K <- J	gwespBB	0.31
Indegree popularity (sqrt)	inPopSqrt	0.048
Outdegree popularity (sqrt)	outPopSqrt	-0.37
Outdegree activity (sqrt)	outActSqrt	-0.0010
Reciprocity x GWESP I -> K -> J	recip x gwespFF	-0.89
Stress ego	egoX	-0.050
Stress alter	alterX	-0.065
Stress similarity	simX	0.55
Change between t1 and t2	rate 1	21.60
Change between t2 and t3	rate 2	19.60
Change between t3 and t4	rate 3	14.50

Note: The parameters that relate to social influence and its interaction with consensus are highlighted gray.

Figure 4 shows the comparison of power estimates between nine scenarios of effect size combinations. The x axis represents the effect size of average similarity effect. The y axis represents the power estimates, where each estimate is based on 100 re-estimated simulated data sets. The three effect sizes of the interaction term are given by the three lines with 95 confidence intervals. As shown, the lowest power is found when the effect size of average similarity equals 3, and the effect size of its interaction with peer variance equals -6. However, this combination (i.e., a small social influence effect with a large moderating effect of peer variance; the latter

double the former) is unusual, and we will not expect that to occur in actual networks. And the power substantially increases if the assumed effect sizes go up. In most of these models, the power is satisfying.

Figure 4. Power of the interaction effect between social influence and peer variance in models with different effect sizes of social influence and its interaction with peer variance



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